

Aditya Madabhushi

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TECHNICAL SKILLS

Programming: Python, Javascript, Java, Golang, SQL

Systems, Tools & Cloud: Docker, Kubernetes, Linux, Git/Github, AWS, GCP, Selenium, Scrapy

Web Development: Node, React, Express, Streamlit, Flask, HTML, CSS, Bootstrap, Tailwind CSS, Material UI

Data Analysis & Viz: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Chart.js, D3.js, Excel

ML, AI & GenAI: TensorFlow, PyTorch, Scikit-learn, NLTK, LangChain, Hugging Face, OpenAI, Groq, Pydantic

Databases: PostgreSQL, MySQL, MongoDB, Neo4J, DynamoDB, SQLite, ChromaDB, Pinecone, Qdrant

EXPERIENCE

Artificial Intelligence Engineer Intern

May 2022 - Aug 2022

Cambridge Technologies, Hyderabad

- Performed Apple Disease Segmentation using an apple leaf dataset and trained a U-Net model with OpenCV-generated masks, achieving 98.2% accuracy in detecting Black Rot.
- Used Google Cloud Vertex AI's AutoML Image Classifier to classify palm trees with 90% accuracy, and applied YOLO for palm tree counting, achieving 85% accuracy through hyperparameter tuning.
- Presented a solution paper to senior leadership, effectively communicating key insights, model performance metrics, and recommendations for implementation.

Data Analyst and Computer Vision Research Assistant

May 2022 - Jan 2023

Gandhi Institute of Technology and Management, Hyderabad

- Developed a custom dataset of 7,000 images (smokers and non-smokers) and performed exploratory data analysis (EDA) to improve data quality and model performance.
- Designed and deployed deep learning models (DenseNet, Xception, Inception, ResNet50) integrated into a voting classifier achieving 94.41% accuracy, and used YOLOv5 for cigarette detection, attaining 93% accuracy.
- Co-authored two research papers: "Smoking Detection using Deep Learning" and "Region Extraction based Approach for Cigarette Usage Classification using Deep Learning."

PROJECTS

AWS-Based Serverless Face Recognition in Video System

- Built a serverless face recognition system that processed 100 videos with an average time of 2.5 seconds per video, ensuring fast and reliable face detection, with user authentication managed via AWS Cognito.
- Developed an AWS Lambda function which splits videos into frames using FFmpeg, then processed each frame with the SSD algorithm which leverages a pre-trained ResNet-34 CNN model and hosted it on Elastic Container Registry (ECR).
- Ensured seamless delivery of results to users via WebSocket API, providing a smooth experience through a React and Tailwind CSS frontend hosted on AWS Amplify.

Course Schedule Management System

- Utilized a MySQL database for storing and retrieving course schedules, enabling efficient data organization, secure handling of enrollment trends, and user authentication with role-based access control and bcrypt encryption.
- Built interactive dashboards using Chart.js, Bootstrap, and React to visualize course availability and enrollment stats, providing tailored views for admins, faculty, and students, enhancing decision-making with insights into course trends and enrollment behavior.

End-to-End Data Engineering Pipeline for Graph Insights

- Developed a data ingestion pipeline which begins with preprocessing 300k parquet files using Python, benchmarked in under 6 minutes.
- Utilized Kafka to stream the processed records to Neo4j deployed on managed Kubernetes, configured using YAML to support large messages.
- Identified busy hubs in the graph using the page rank algorithm, with the database read latency recorded under 2 seconds for a graph containing around 50 nodes.

Advanced Depression Detection with Enhanced Feature Engineering

- Designed and implemented a Voting Classifier using Random Forest, KNN, and Logistic Regression to predict depression from social network data, incorporating text preprocessing with NLTK and spaCy.
- Engineered a feature pipeline integrating numerical and text features with TF-IDF, including emoticon-to-word conversion for enhanced semantic analysis, achieving a 93.1% F1-Score.

EDUCATION

M.S. Computer Science

May 2025

Arizona State University, Tempe

4.0 GPA

- Coursework:** Statistical Machine Learning, Foundation of Algorithms, Data Processing at Scale, Cloud Computing, Distributed Software Development, Generative AI

B.Tech. Computer Science

May 2023

Gandhi Institute of Technology and Management, Hyderabad

3.65 GPA

- Coursework:** Database Systems Management, Big Data Analytics, Agile Software Development, Operating System